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ORIGINAL PAPER

CHDPL-Net: a lightweight network for Chinese herbal decoction pieces detection

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Abstract

To advance the integration of traditional Chinese medicine (TCM) with next-generation information technologies, the intelligent identification of Chinese herbal decoction pieces (CHDP) has become a crucial research direction. However, the performance of current algorithms remains unsatisfactory. To address this, we have constructed a diverse CHDP dataset and proposed a lightweight network for CHDP detection, named CHDPL-Net. Based on YOLOv8, this model introduces a new network scaling factor to reduce redundant channels in deep feature maps and optimizes the Neck and Head structures to better accommodate CHDP detection, which primarily involves medium and large targets. Additionally, a newly designed downsampling module, RDown, replaces conventional downsampling methods to reduce computational overhead, while the adopted upsampling module, DySample, significantly enhances the recovery of detailed features. To further improve lightweight performance, we apply GhostConv to optimize the SPPF and C2F modules and incorporate a novel attention mechanism, EHA, which makes the model more sensitive to color and texture information, mitigating the performance degradation caused by lightweight design. Ultimately, CHDPL-Net achieved excellent results with only 31.9% of the Parameters and 30.6% of the FLOPs compared to YOLOv8, obtaining mAP_{50} and $mAP_{50:95}$ scores of 98.2% and 95.4%, respectively, with only a 0.8% performance drop. This demonstrates that the model can meet practical detection needs to a certain extent.

Keywords Chinese herbal decoction pieces · Object detect · YOLOv8 · Lightweight model

1 Introduction

Chinese herbal decoction pieces (CHDP), defined as processed medicinal materials prepared under the guidance of traditional Chinese medicine (TCM) theory for direct clinical dispensing or pharmaceutical production, serve as the fundamental therapeutic agents in TCM practice. With the deepening implementation of the Digital China strategy, the TCM industry is accelerating its integration with next-generation information technologies. At the national policy level, the "14th Five-Year Plan for TCM Development" explicitly advocates for promoting inclusive application of digitalized TCM services, identifying innovative service models such as shared herbal pharmacies and smart pharmacies as key development objectives. Whereas as the core pharmaceutical carrier in TCM clinical practice, CHDP traditional identification predominantly relies on experienced pharmacists' morphological characteristics evaluation, which presents limitations including strong subjectivity, low standardization, and challenges in large-scale replication. Moreover, the cultivation cycle for senior pharmacists typically spans 5-8 years and involves significant training costs and resources. Against this background, intelligent identification technology for CHDP based on computer vision has become an important research domain.

Early CHDP intelligent identification technologies were primarily based on image processing techniques and machine

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learning algorithms (Tao et al. 2014; Kan et al. 2017; Xie et al. 2018; Chen et al. 2020). Typically, these methods involved feature extraction using various image processing techniques, followed by classification with algorithms such as KNN (Peterson 2009), SVM (Cortes and Vapnik 1995), etc. However, algorithms relying on low-level physical features often perform inadequately in complex scenarios. With the rapid development of deep learning theories and related software and hardware, many researchers have shifted their focus to deep learning algorithms, including Faster R-CNN (Ren et al. 2015), SSD (Liu et al. 2016), YOLO series, among others. This advancement not only moves CHDP identification beyond single-object classification but also enhances its performance in complex scenarios, further improving the feasibility of deploying CHDP identification technology and potentially replacing traditional manual identification methods.

Despite some progress in CHDP identification, the performance of existing algorithms remains unsatisfactory due to several reasons: (1) There is a lack of high-quality, publicly available CHDP datasets. Many studies use images of varying quality, some of which are sourced from web scraping. These images may not align with real-world application scenarios or have been artistically modified, as shown in Fig. 1. Excessive irrelevant environmental noise makes it difficult for models to learn key CHDP features, thereby degrading recognition performance. (2) To enhance the recognition capabilities of models, numerous studies have increased model depth to extract more feature information. However, this approach significantly increases the number of parameters and computational complexity. Given that CHDP identification is often applied in contexts such as traditional Chinese medicine market transactions, retail in herbal shops, and public medication usage, where deployment on portable mobile devices is preferable, achieving a balance between recognition performance and lightweight design remains a significant challenge.

To address these challenges, we manually captured CHDP images and performed data labeling. Additionally, considering the generalization ability, parameter efficiency and detection speed of the model, we chooses to improve and optimize on the basis of YOLOv8 (Varghese and Sambath 2024), and propose a lightweight network named CHDPL-Net for CHDP detection. The main improvements are as follows:

(1) We propose YOLOv8t by adjusting the network scaling factor of YOLOv8, making it smaller than the smallest YOLOv8n. This network reduces channel redundancy in deeper layers while avoiding adverse effects on feature extraction.
(2) Given that CHDP identification primarily involves medium and large targets, we removed the P3 layer responsible for small target detection from the Neck and Head sections and added an additional downsampling layer to retain multi-granularity information, aiding in the detection of medium and large targets. This significantly reduced the number of parameters and computational load while avoiding interference from small target detection head.
(3) To reduce the burden on the downsampling layer, we proposed a reparameterizable downsampling layer (RDown) to replace the CBS convolution module. This approach reduces the number of parameters and computational load while enhancing the diversity of feature extraction and the model's convergence ability, thereby improving computational efficiency and detection performance.
(4) Additionally, we utilized DySample (Liu et al. 2023) to replace the traditional nearest-neighbor interpolation method, enabling the upsampling layer to autonomously learn and possess spatial adaptability while only increasing a small number of parameters. This approach enhances the recovery of CHDP's detailed textures.

Fig. 1 Images from the web

(5) To further reduce the number of parameters and computational load, we applied GhostConv (Han et al. 2020) to lightweight the SPPF, Bottleneck, and C2F modules. To mitigate the performance degradation caused by these modifications, we proposed an Efficient Hybrid Attention (EHA) mechanism based on the color and texture sensitivity of CHDP. By integrating EHA into the C2F module, we enhanced feature extraction capabilities, enabling the model to better learn the color and texture features of CHDP.

The rest of this paper is organized as follows: Section 2 reviews the related work on CHDP identification and model lightweighting. Section 3 describes the constructed datasets and introduces the overall structure of proposed method. Section 4 describes the experimental settings and results, including ablation experiments, comparative experiments and robustness experiments. Section 5 concludes this paper.

2 Related works

2.1 CHDP identification

CHDP intelligent identification methods include traditional machine learning methods and deep learning methods. Traditional machine learning methods primarily utilize image processing techniques to extract features from images and use classifiers for classification (Table 1). Tao et al. (2014) extracted features using the Gray-Level Co-occurrence Matrix and the Gray-Level Gradient Matrix, and established classification models using the Naive Bayes method and the Backpropagation Neural Network method. Kan et al. (2017) extracted ten shape features and five texture features, and used an SVM for classification. This approach, by integrating both shape and texture features, significantly enhanced the recognition accuracy. Xie et al. (2018) first used an improved watershed algorithm for image segmentation and then extracted color features using color histograms. Finally, they utilized an SVM for classification. Chen et al. (2020) constructed a color template that is invariant to size and rotation, then integrated color features from two observation angles, calculating the correlation with the template. This approach improved robustness against variations in sample shape, sampling location, and color changes. Although these methods based on low-level physical features can perform recognition to some extent, they are susceptible to interference from environmental factors and CHDP quality, leading to lower reliability, and the recognition process is relatively complex.

In contrast, some other studies (Sun and Qian 2017; WU et al. 2020; LIU et al. 2021) have demonstrated the potential of deep learning in the field of CHDP identification. Compared to traditional machine learning methods, deep learning methods extract features that are not low-level physical characteristics but rather high-level semantic information and employ an end-to-end recognition approach. As a result, these methods show significant improvements in both accuracy and speed and are no longer limited to single-target classification problems. This has attracted many researchers to shift their focus towards deep learning models (Table 2). Hu et al. (2020) leveraged the concept of multi-task learning, combining neural networks with traditional features to improve recognition accuracy and stability. CHEN and ZOU (2021) introduced the Wasserstein distance into Generative Adversarial Network (GAN) and added conditional parameters to enhance recognition accuracy. Miao et al. (2023) integrated ConvNeXt with the ACMix network to improve feature extraction performance. These studies primarily aimed to enhance model recognition accuracy. To better deploy these models in practical applications, some researchers have also focused on model lightweighting. Gang et al. (2023) designed Group Convolution Inverted Residual using bottleneck structures and depthwise separable convolutions to reduce parameter counts, and drew inspiration

Table 1 Related works of traditional machine learning methods for CHDP Identification

Reference	Advantages	Disadvantages
Tao et al. (2014)	It was simple and highly interpretable.	The identification process was complex, leading to low efficiency.
Kan et al. (2017)	It integrated both shape and texture features.	The SVM method is relatively simple in structure, which limits its capability for deeply fusing shape and texture features.
Xie et al. (2018)	Effective segmentation improved feature accuracy.	The method relied solely on color features, resulting in limited robustness in complex environments.
Chen et al. (2020)	It effectively reduced the impact of observation angles on the recognition results.	This effect was limited to color features.

Table 2 Related works of deep learning methods for CHDP Identification

Reference	Advantages	Disadvantages
Hu et al. (2020)	It combined neural networks with traditional features.	The model was relatively complex, leading to low efficiency.
CHEN and ZOU (2021)	It improved accuracy by leveraging GAN and Wasserstein distance.	It showed relatively lower performance in distinguishing similar categories.
Miao et al. (2023)	It combined ConvNeXt with the ACMix network to improve feature extraction performance.	The model complexity increases significantly.
Gang et al. (2023)	The optimized structure reduces the number of parameters and is designed with MDCA to improve perception.	It was limited to single-object classification, and the MDCA was insensitive to texture features.
Li et al. (2024)	It adjusted the convolutional kernel sizes in AlexNet to reduce parameter count and incorporated the SE to improve recognition accuracy.	The recognition accuracy was low, and the SE was also insensitive to texture features.

from coordinate attention (CA) and triplet attention (TA) to design multidimensional channel attention (MDCA), thereby enhancing perceptual capabilities. Li et al. (2024) adjusted the convolutional kernels of AlexNet to reduce parameter counts, replaced normalization layers and activation functions to enhance network expressiveness, and incorporated squeeze-and-excitation networks (SE) to boost recognition accuracy. These innovations make low-cost identification of CHDP feasible, but there is still room for further optimization.

2.2 Model lightweighting

Model lightweighting is a critical pathway for overcoming computational bottlenecks and achieving practical deployment. The main lightweighting techniques include knowledge distillation, parameter pruning, quantization compression, low-rank decomposition, and lightweight architecture design. Among these, lightweight architecture design methods, represented by depthwise separable convolutions (DWSConv) (Chollet 2017), inverted residual structures, and attention mechanism optimizations, have become a key focus of research due to their ability to strike a balance between computational efficiency and feature representation capabilities.

Within the realm of lightweight architecture design, the YOLO series, as a representative of single-stage object detection algorithms, has been widely used by many researchers as a base model for corresponding improvements to better suit specific tasks (Table 3). Tang et al. (2024) designed parallel dilated residual modules and a selective bidirectional diffusion pyramid network to obtain richer pedestrian features, and added a detail feature layer to capture information from small targets. However, these operations inadvertently increased the computational load of the model. Additionally, constructing a lightweight shared detection head adversely affects the model's multi-scale detection performance. Liu et al. (2024) adjusted the scaling factor of YOLOv8 to halve the number of feature map channels, replaced the original C2F feature extraction module with PDWFasterNet, and used lightweight DWSConv instead of conventional convolutions for downsampling. Additionally, they introduced EIoU loss to mitigate the impact of lightweight improvements on detection accuracy. However, indiscriminately halving

Table 3 Related works of YOLO lightweighting

Reference	Advantages	Disadvantages
Tang et al. (2024)	It captures richer pedestrian features and obtains more detailed information about small targets.	It increased the computational burden of the model due to some modules, and the lightweight shared detection head was not conducive to multi-scale detection.
Liu et al. (2024)	It adjusted the scaling factor and utilized PDWFasterNet along with DWSConv to achieve lightweighting.	Indiscriminately halving the number of feature map channels had a significant impact on model performance.
Wu et al. (2024)	The integration of CBAM, CARAFE, and WIOU techniques significantly improved detection accuracy.	This model had relatively large Parameters and FLOPs, reaching 10.46M and 25.4G, respectively.
Feng et al. (2024)	While decreasing the parameter count, the model placed more emphasis on detecting small objects.	The adoption of CA, AMFF, and Dynamic Head increased computational cost, leading to 41.8G FLOPs.
Dang et al. (2025)	It significantly reduced the number of parameters by employing LFEM, LDown, and shared convolutional blocks.	The parameter efficiency of LFEM is decreased owing to the lack of a bottleneck structure.

the number of feature map channels significantly affects model performance. Wu et al. (2024) enhanced the focus on apples by incorporating the convolutional block attention module (CBAM) and the CARAFE upsampling operator, and reduced the computational load of the Neck using GSConv and VoV-GSCSP. Feng et al. (2024) removed the P5 layer and added a P2 layer in the Neck, while feeding both the P3 and P2 layers into the detection head, making the model more focused on small targets and reducing the number of parameters. Additionally, they integrated CA attention mechanism into C2F to enhance the model's focus on critical information. Dang et al. (2025) utilized a Lightweight Feature Extraction Module (LFEM) and a sparse optimization module (LDown) to replace parts of the C2F and downsampling convolutions, and introduced shared convolution blocks in the detection head to reduce the number of parameters. However, LFEM may reduce parameter efficiency, by discarding the bottleneck structure. These innovations have significantly improved the computational efficiency of models on resource-constrained edge devices, but achieving a balance between recognition performance and lightweight design remains challenging.

3 Materials and methods

3.1 Dataset construction

Training deep learning models requires high-quality datasets. This study constructs the CHDP4K dataset for detecting CHDP. We used mobile phones to capture images of 38 commonly used CHDPs placed on a desktop, resulting in a total of 4,004 images, as shown in Fig. 2. These images include two resolutions (1228×1228 and 2992×2992), two desktop backgrounds (no interference and wooden texture interference), various lighting conditions, and different shooting heights to simulate realistic CHDP usage scenarios. Furthermore, we constructed a more challenging dataset named CHDP5H in the Robustness experiments Section 4.5, which includes visually similar CHDPs and more complex scenarios.

During the data annotation phase, we utilized the LabelImg software (https://github.com/HumanSignal/LabelImg) for our operations. This tool helped us efficiently label the position and category information of each object in every image. Our strategy for data augmentation is to implement it during the training phase rather than the data preparation phase. Specifically, the data augmentation methods we employed included adjustments to hue, saturation, and brightness, translation transformations, scaling transformations, horizontal flipping, and mosaic data augmentation (Chollet 2017). Detailed parameter settings for each augmentation method are provided in Table 4, and the enhancement effects, which include nine augmented images, are demonstrated in Fig. 3.

For dataset division, considering that no hyperparameter tuning was performed, we split the dataset into training and validation sets at an 8:2 ratio. This allowed us to fully utilize the dataset for training and validation while ensuring that the validation set was used for testing model performance. By doing so, we maximized the use of the dataset to improve the overall performance of the model.

3.2 YOLOv8

The structure of YOLOv8 is shown in Fig. 4 and includes the following four parts. Compared with other models in YOLO series, the overall structure of YOLOv8 is simple and efficient, and thus it was selected as the base model for this study.

Fig. 2 Images from the CHDP4K

Table 4 Details of data augmentation parameters

Parameter	Value	Description
hsv_h	0.015	Hue is adjusted between 0.985 and 1.015 of the original value
hsv_s	0.7	Saturation is adjusted between 0.3 and 1.7 of the original value
hsv_v	0.4	Brightness is adjusted between 0.6 and 1.4 of the original value
translate	0.1	Horizontal and vertical translation amplitude of 0.1 relative to image size
scale	0.5	Image scale factor ranges from0.5 to 1.5
fliplr	0.5	50% probability of performing horizontal flip
mosaic	4	Randomly crop patches from four images and stitch them together

(1) **Feature Extraction Module**: Replacing the C3 module in YOLOv5 with C2F provides richer gradient flow, enhancing feature representation capabilities while maintaining a lightweight structure.

(2) **Decoupled Head Design**: The head part adopts a decoupled head design, isolating parameters for classification and localization tasks to avoid gradient interference, enabling both tasks to converge efficiently and simultaneously.

(3) **Anchor-Free Mechanism** (Zhang et al. 2020): By adopting an Anchor-Free approach to replace the anchor-based mechanism, it is no longer necessary to meticulously design and adjust the sizes and quantities of anchors. This not only simplifies the network architecture but also accelerates the post-processing of non-maximum suppression (Neubeck and Van Gool 2006), thereby reducing the computational load.

(4) **Task-Aligned Learning** (Feng et al. 2021): Implementing task-aligned learning sample allocation strategies helps the model focus better on high-quality samples that are crucial for both classification and localization, thereby improving overall model performance.

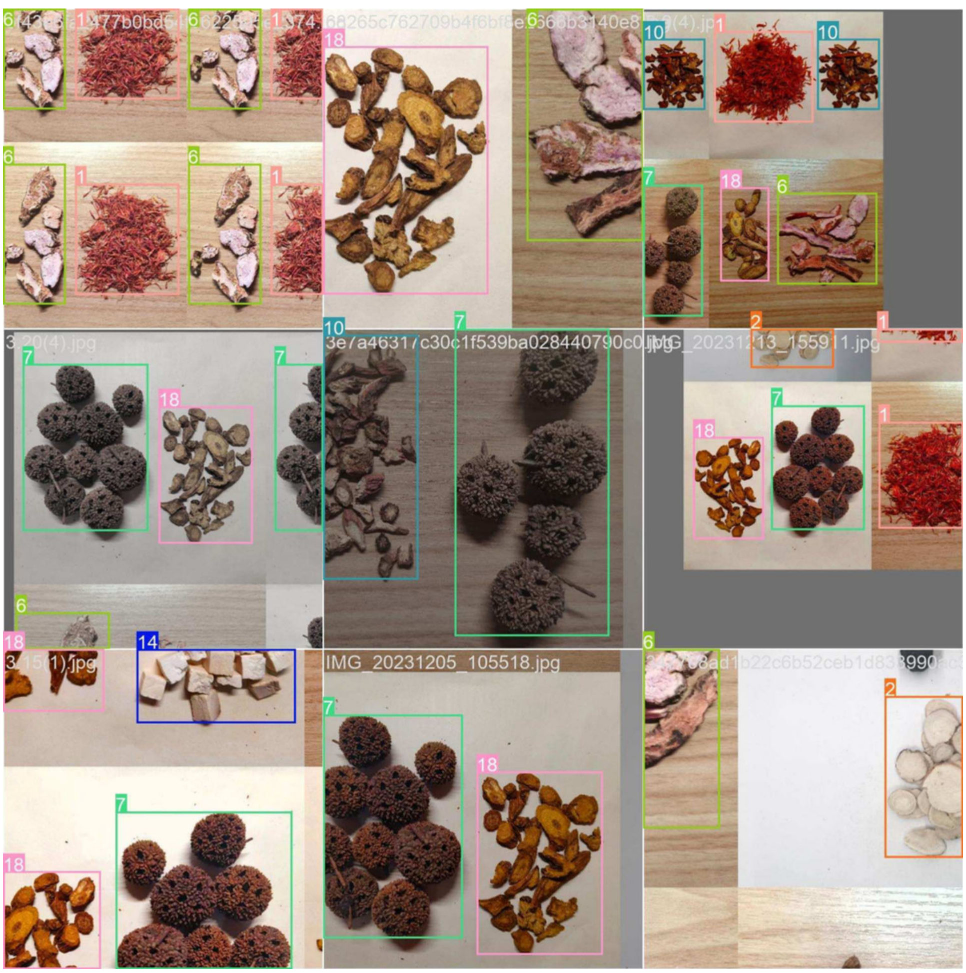


Fig. 3 Images after data augmentation

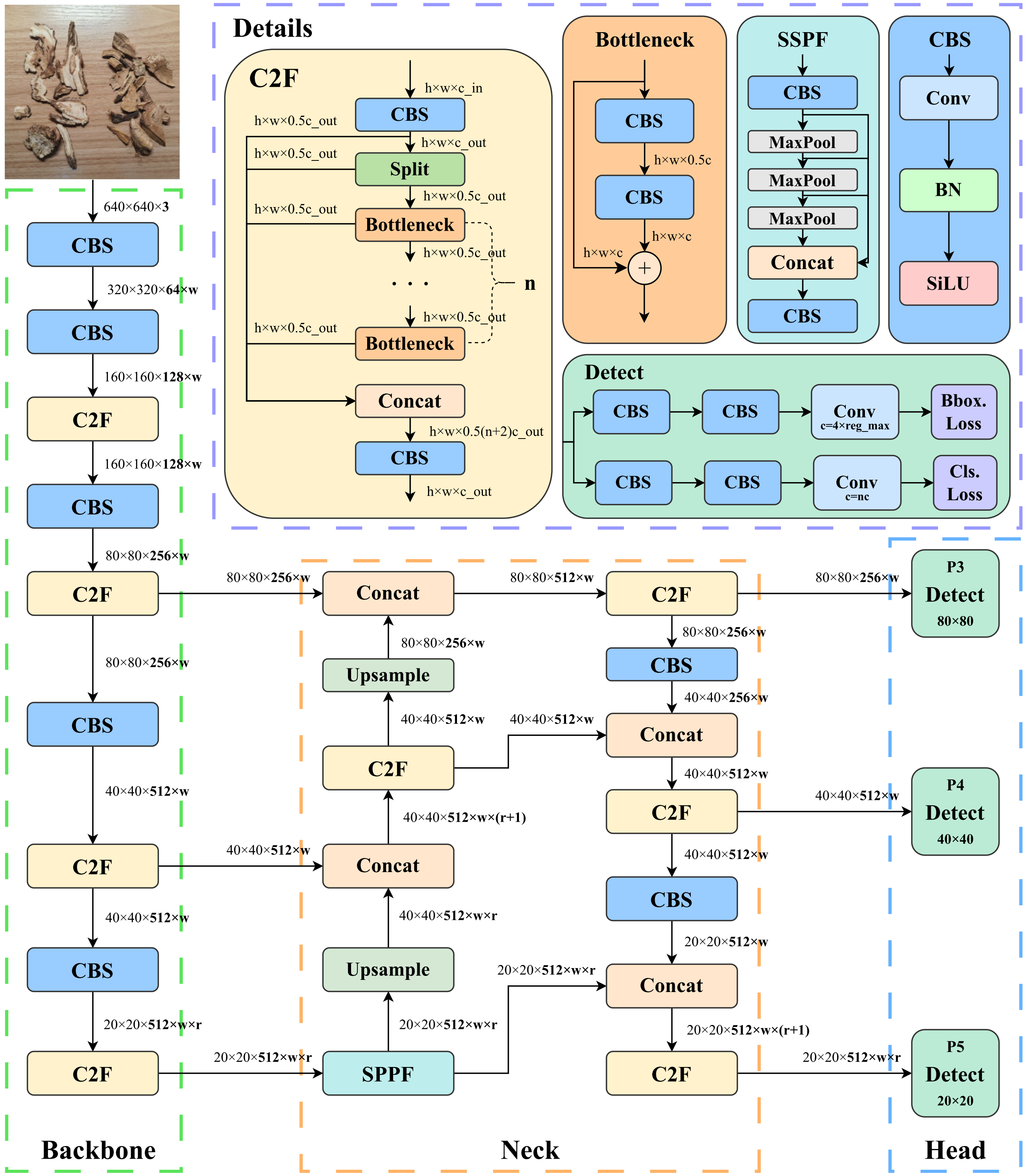


Fig. 4 Structure of YOLOv8

3.3 CHDPL-Net

The structure of CHDPL-Net is shown in Fig. 5. In the following subsections, we will provide a detailed description of each improvement.

3.3.1 YOLOv8t

To meet the needs of different application scenarios, YOLOv8 provides five different network configurations by adjusting the network scaling factors. Among these factors, w

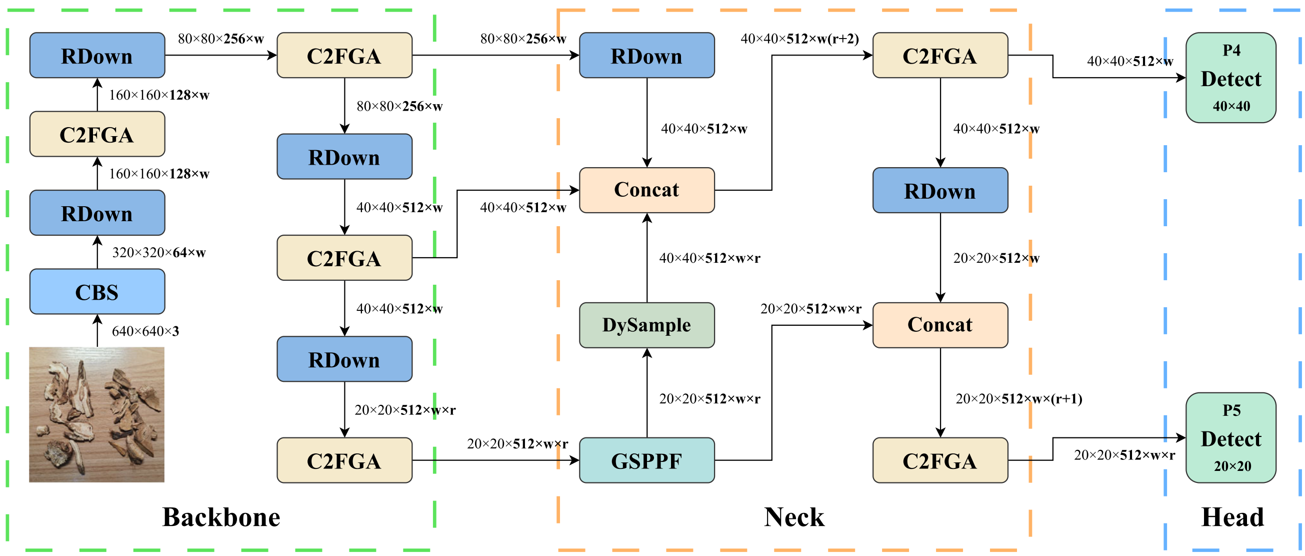


Fig. 5 Structure of CHDPL-Net

(widen_factor) is used to adjust the network width, i.e., the number of channels per layer; d (deepen_factor) adjusts the network depth, specifically reflected in the number of Bottleneck uses within the C2F module; r (ratio) assists w in controlling the number of channels in deep feature maps, making the model more adaptable to various tasks. Higher scaling factors are expected to achieve better performance but will significantly increase the model's parameters and computational load, potentially leading to overfitting.

Considering that apple detection only involves one category, Liu et al. (2024) proposed YOLOv8p by reducing w to half of the smallest YOLOv8n, meaning the number of channels in all layers was halved. This is a rather aggressive modification. In CHDP identification, given the involvement of multiple categories with high similarity between some classes, we adjusted r to half of YOLOv8n, meaning only the number of channels in deep feature maps was halved. This conservative adjustment aims to reduce the number of parameters and computational load while minimizing the impact on the model. Different network scaling factors are shown in Table 5.

Table 5 Details of scaling factors of YOLOv8

Model	w	d	r
YOLOv8n	0.25	0.33	2.0
YOLOv8s	0.50	0.33	2.0
YOLOv8m	0.75	0.67	1.5
YOLOv8l	1.00	1.00	1.0
YOLOv8x	1.25	1.00	1.0
YOLOv8p	0.125	0.33	2.0
YOLOv8t (Ours)	0.25	0.33	1.0

3.3.2 Improved neck and head structure

In YOLOv8, the network was downsampled a total of five times to generate five sets of feature maps with different sizes, 320 × 320 (P1), 160 × 160 (P2), 80 × 80 (P3), 40 × 40 (P4) and 20 × 20 (P5) pixel sizes, and PANet was used to perform multi-scale feature fusion from the P3 to the P5 layer, which was finally sent to the corresponding detection head for detection. Specifically, P3 is used for detecting small targets, P4 for medium targets, and P5 for large targets.

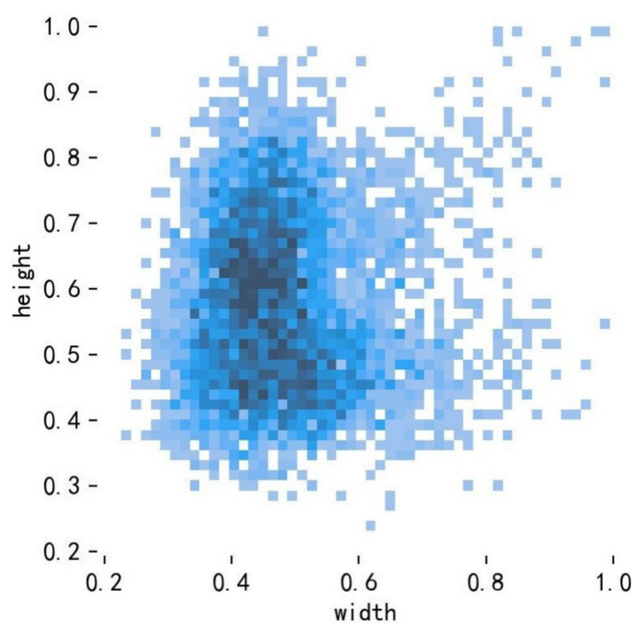


Fig. 6 Target aspect ratio distribution

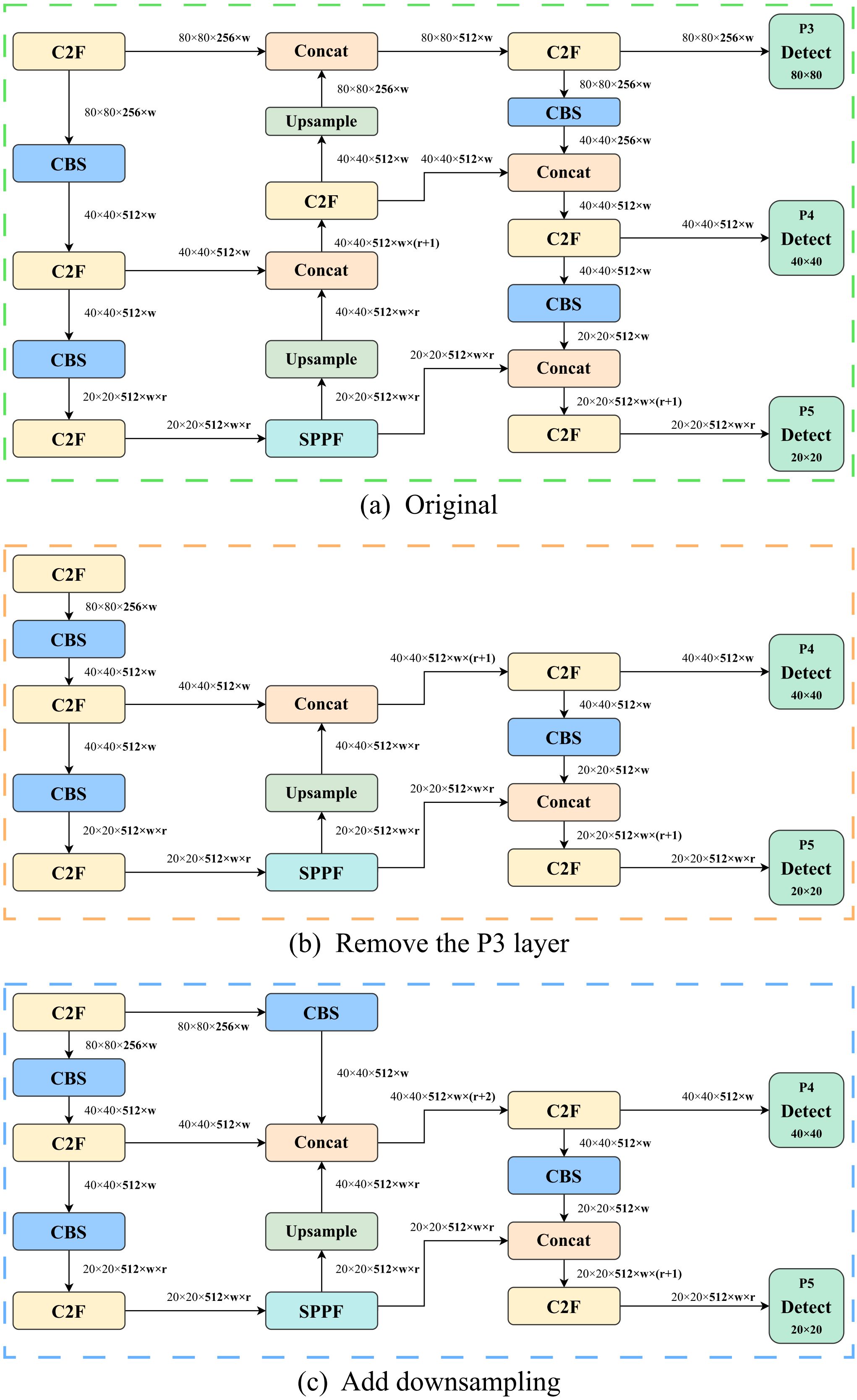
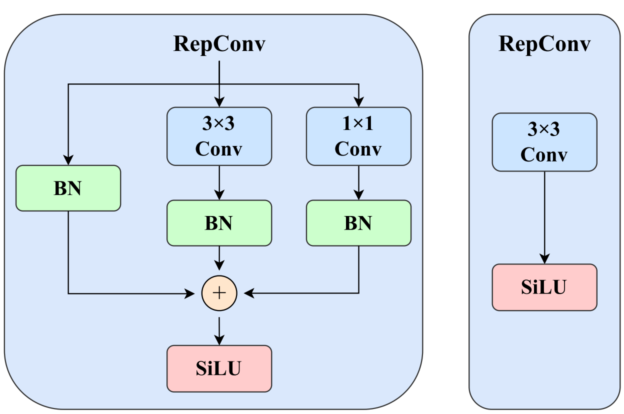


Fig. 7 The neck and head structure optimization

Given that CHDP recognition involves close-range imaging where mainly medium and large targets are considered, as illustrated by the target-to-image size ratios shown in Fig. 6, which are concentrated between 0.3 and 0.9. Based on this, this study proposes an optimization strategy that removes the P3 layer from the Neck and Head to reduce model parameters and avoid the influence of small target detection head. However, completely removing the P3 layer disrupts the multi-scale feature fusion process in the Neck. To ensure that P4 and P5 layers can still utilize fine-grained information, we introduce an additional downsampling layer to incorporate P3's information into the multi-scale feature fusion, as depicted in Fig. 7. This improvement not only significantly reduces the number of model parameters but also enhances the detection efficiency for medium and large targets.

3.3.3 RDown

In YOLOv8, the downsampling operation typically employs the standard CBS module, which requires a significant amount of parameters and computational resources, making lightweight deployment challenging. To address the issues, we propose a design that combines feature splitting with reparameterization techniques. Feature splitting divides the input feature maps into two sub-feature maps for separate processing to reduce the number of parameters and computational load. One sub-feature map is processed using maximum pooling to capture salient features, while the other sub-feature map is processed using average pooling to retain global information. The reparameterization technique enhances feature extraction and convergence during training by introducing additional Conv paths. During inference, these paths are reparameterized into a single Conv to ensure no increase in parameters or computational load, as illustrated in Fig. 8.



(a) Training (b) Inference

Fig. 8 Reparameterization technique

First, the Conv and BN in each of the three branches are fused separately into new Conv, as shown in (1), (2), (3). The left branch can be considered as the fusion of a 1×1 Conv with weights W all set to 1 and bias b set to 0, and a BN.

$$Conv(x) = W(x) + b \tag{1}$$

$$BN(x) = \gamma * \frac{x - mean}{\sqrt{var}} + \beta \tag{2}$$

$$\begin{aligned} BN(Conv(x)) &= \gamma * \frac{Conv(x) - mean}{\sqrt{var}} + \beta \\ &= \gamma * \frac{W(x) + b - mean}{\sqrt{var}} + \beta \\ &= \frac{\gamma * W(x)}{\sqrt{var}} + (\frac{\gamma * (b - mean)}{\sqrt{var}} + \beta) \\ &= W'(x) + b' \end{aligned} \tag{3}$$

where W' and b' represent the weights and bias of the new Conv, respectively. Then, the new Convs ($Conv'_{0\times0}$, $Conv'_{3\times3}$, $Conv'_{1\times1}$) from the three branches are fused into a final Conv ($Conv''_{3\times3}$), as shown in (4).

$$\begin{aligned} Conv''_{3\times3}(x) &= Conv'_{0\times0}(x) + Conv'_{3\times3}(x) + Conv'_{1\times1}(x) \\ &= W'_{0\times0}(x) + W'_{3\times3}(x) + W'_{1\times1}(x) \\ &\quad + b'_{0\times0} + b'_{3\times3} + b'_{1\times1} \\ &= W''(x) + b'' \end{aligned} \tag{4}$$

The structure of RDown is shown in Fig. 9. It is used to replace all downsampling layers in the network except for the first one. RDown captures salient features and retains global information while reducing the number of parameters and computational load.

3.3.4 DySample

In the feature fusion network of YOLOv8, the upsampling module used the nearest-neighbor interpolation method, which fills in pixels by selecting the nearest pixel values. However, this method overlooks the intrinsic relationships between pixels, affecting the recovery of detail in feature maps. This has a particularly significant impact on CHDP that rely heavily on texture information for identification. To address this issue, we introduce DySample as a replacement for nearest-neighbor interpolation to enhance performance.

The structure of DySample is illustrated in Fig. 10. For the input feature X, the Sampling Point Generator first generates a sampling set S, which contains g groups of coordinates. Then, S and X are divided into g groups and sampled according to the coordinates. Inside the Sampling Point Generator, X passes through two linear branches: one branch generates

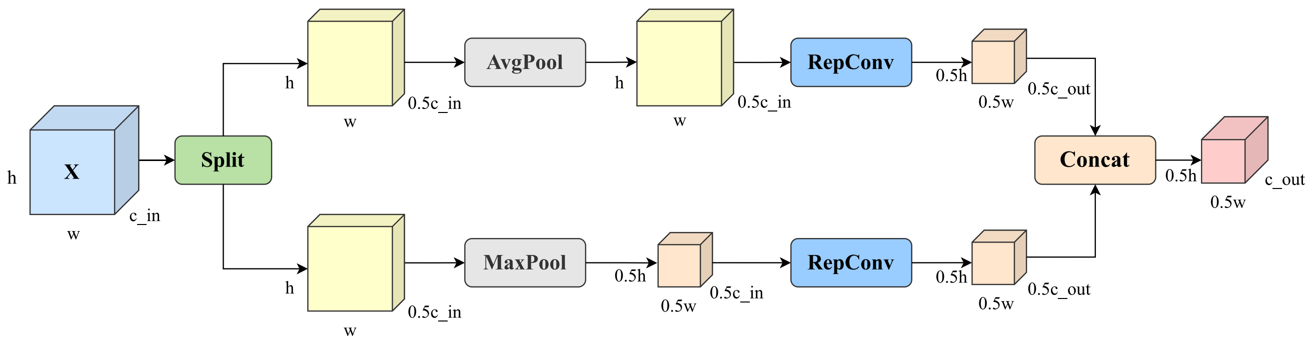


Fig. 9 Structure of RDown

factors within the range of −0.5 to 0.5, which are multiplied with the results from the other branch to obtain offsets. These offsets are then processed through a pixel shuffle operation and added to the initial positions of each point to derive the final sampling coordinates S.

DySample employs a simple yet effective method to generate content-aware upsampling results for each point. For CHDP that depend on texture information for identification, this approach better restores detail information, thereby enhancing model performance. Moreover, the additional parameters and computational load introduced by DySample are negligible compared to the overall model.

3.3.5 Ghost model

Considering that the C2F module primarily uses bottleneck structures and skip connections to reduce parameters and computational load while providing richer gradient flow, its main operator remains the CBS module, which still consumes a significant amount of parameters and computations.

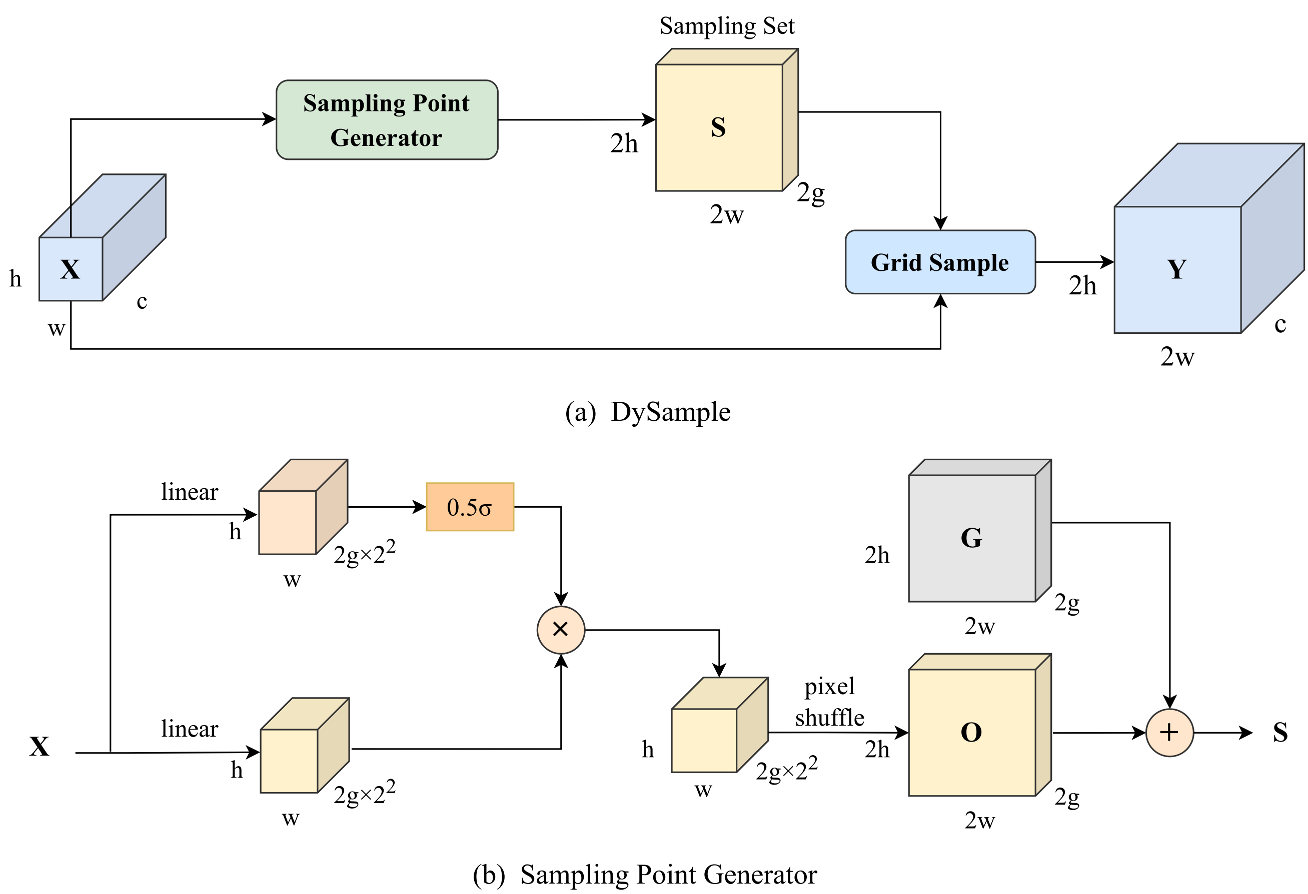


Fig. 10 Structure of DySample

To address this issue, we propose a new bottleneck module called GBottleneck, as shown in Fig. 11.

In GBottleneck, we replace the CBS module with GhostConv to reduce parameters and computational load. GhostConv addresses the redundancy problem in feature maps by proposing a structure that generates a large number of feature maps with minimal computation. Its impact on performance is significantly lower than that of DWSConv, as illustrated in Fig. 12. Initially, a regular convolution generates feature maps with half the number of channels to reduce parameters and computational load. Then, an inexpensive operation (depthwise convolution) generates the other half of the feature maps. Finally, the two sets of feature maps are concatenated. The ratios of parameters (r_p) and computational load (r_c) for the GhostConv compared to the CBS are approximately $\frac{1}{2}$, as shown in (5), (6).

$$r_p = \frac{c_{in} * 0.5c_{out} * k^2 + 0.5c_{out} * k^2}{c_{in} * c_{out} * k^2} = \frac{c_{in} * 0.5 + 0.5}{c_{in}} \approx \frac{1}{2} \tag{5}$$

$$r_c = \frac{c_{in} * 0.5c_{out} * h * w * k^2 + 0.5c_{out} * h * w * k^2}{c_{in} * c_{out} * h * w * k^2} = \frac{c_{in} * 0.5 + 0.5}{c_{in}} \approx \frac{1}{2} \tag{6}$$

where h and w represent the height and width of the feature map, c_{in} and c_{out} denote the number of input and output channels, respectively, and k stands for the kernel size of the convolution.

We further propose C2FGA as a replacement for C2F based on GBottleneck, as shown in Fig. 13. First, we replace the CBS and Bottleneck in C2F with GhostConv and GBottleneck to reduce parameters and computational load. Secondly, it is observed that each Bottleneck has its own skip connection, but in C2F, the output of each Bottleneck is still connected to the final CBS, which is redundant. Therefore, we choose to retain only the connection from the last layer of GBottleneck, reducing the input channels of the final GhostConv layer to further minimize parameters and computational load. Finally, we incorporate an EHA attention module (introduced in Section 3.3.6) at the end to mitigate performance degradation caused by lightweight improvements.

Additionally, the SPPF module uses 11 convolutions at the beginning and end, solely for adjusting the number of channels, thereby reducing the computational load of intermediate maximum pooling and the parameters and computations involved in the final fusion. We propose GSSPF, which uses GhostConv to achieve this functionality, resulting in fewer parameters and computational load, as illustrated in Fig. 14.

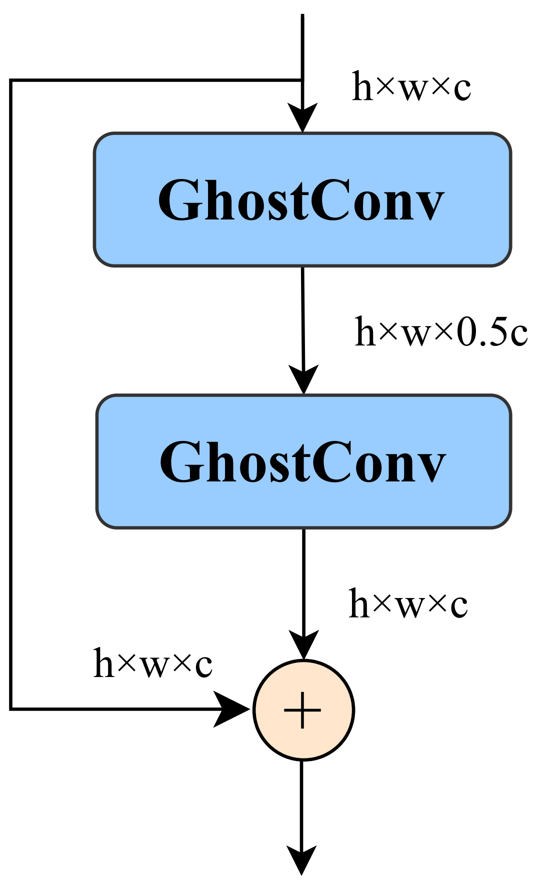


Fig. 11 Structure of GBottleneck

3.3.6 Efficient hybrid attention

To address the issue of fine-grained feature information loss during lightweight modifications, we propose a novel attention module named EHA, based on the color gradient sensitivity and complex texture structure characteristics of CHDP. The design inspiration comes from the crucial role of color and texture in CHDP identification. According to

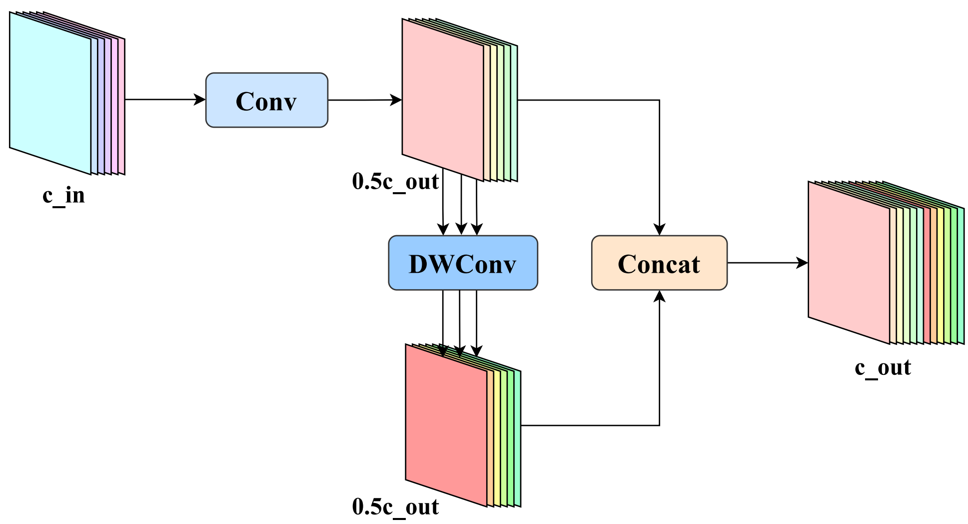


Fig. 12 Structure of GhostConv

TCM principles, color attributes (such as the reddish-brown color of Mu Dan Pi and the yellowish-white color of Bai Xian Pi) and surface textures (for example, the stem of Ze Lan is square-cylindrical with shallow longitudinal grooves on all four sides, while the stem of Pei Lan is cylindrical with distinct longitudinal ridges) are not only visual features but also direct proxies for pharmacological authenticity. This module is integrated into the tail end of C2FGA to enhance the model's perception capabilities for color and texture features. This module combines the advantages of ECA and SA, employing a parallel dual-branch structure to achieve complementary enhancement of channel and spatial information, thereby explicitly amplifying these distinguishing characteristics. It is also designed as a lightweight component, as illustrated in Fig. 15.

The channel attention branch adopts a dynamic kernel width mechanism, utilizing a one-dimensional convolution of size K to fuse channel features after global average pooling, followed by a Sigmoid activation function to generate

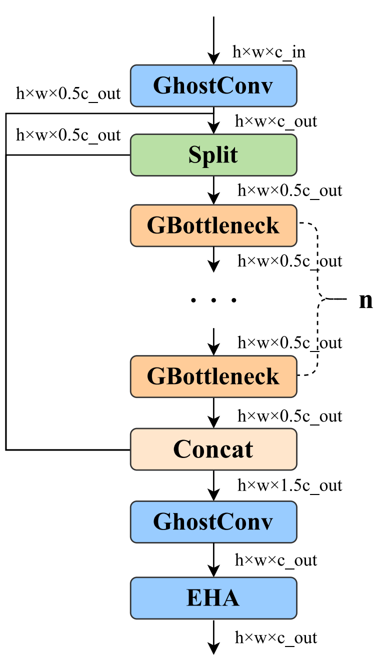


Fig. 13 Structure of C2FGA

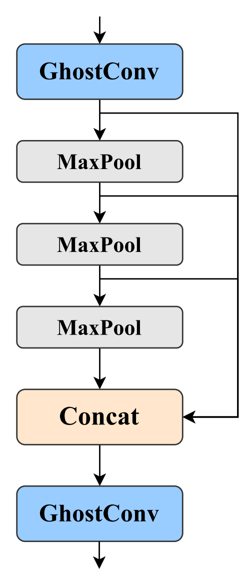


Fig. 14 Structure of GSSPF

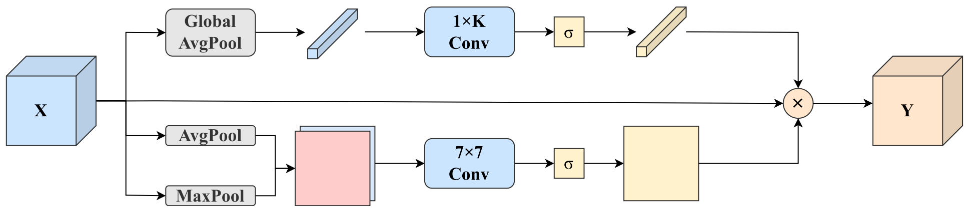


Fig. 15 Structure of EHA

channel attention. Compared to the SE attention mechanism that uses two fully connected layers to fuse channel information, the one-dimensional convolution achieves this with fewer parameters and computational load. The calculation method for K is shown in (7).

$$k = \left\lceil \frac{\log_2(C_{in}) + b}{\gamma} \right\rceil_{\text{odd}} \tag{7}$$

where C_{in} represents the number of input channels, and b and γ are hyperparameters set to 1 and 2, respectively. The notation $\lceil\cdot\rceil_{\text{odd}}$ denotes rounding up to the nearest odd number. This design allows for dynamic adjustment of the kernel width based on the complexity of the input feature maps, thereby enhancing the model's adaptability and efficiency.

The spatial attention branch first performs average pooling and maximum pooling operations to retain global information and extract prominent texture features. It then fuses these results using a 7×7 convolution, followed by a Sigmoid activation function to generate spatial attention. This design effectively captures spatial dependencies within the input feature maps, thereby enhancing the model's ability to perceive detailed textures.

Finally, the generated channel attention and spatial attention are jointly applied to the input feature maps. This ensures enhanced representation capabilities for color and texture features while maintaining computational efficiency, thus improving the accuracy and robustness of CHDP detection.

4 Experiments

4.1 Experimental setting

The operating system used in this study is Linux Ubuntu 20.04, with an NVIDIA GeForce RTX 4060 Ti GPU. The environment is configured with CUDA 12.7, torch 2.5.1, torchvision 0.20.1, and Python 3.9.13. The CHDP4K dataset is split into training and validation sets at a ratio of 8:2, with the validation set used for testing. During training, the input image size is set to 640×640, and the batch size is 8. AdamW is used as the optimizer with an initial learning rate of 0.000238 (calculated as $\frac{0.01}{(4+nc)}$, where nc is the number of classes), following the default configuration in YOLOv8. A weight decay of 0.0005 is applied to prevent overfitting. Over the course of 100 epochs, the learning rate linearly decays to 0.01 times its initial value. To ensure stability during the initial stages of training, a warm-up phase of 3 epochs is employed. Detailed training parameters are summarized in Table 6.

4.2 Evaluation metrics

This study evaluates the model from two aspects: lightweight metrics and performance metrics. Lightweight metrics include the number of model parameters (Parameters) and the computational load per forward pass (FLOPs), which are used to measure the model size and computational burden, respectively. Furthermore, inference time (Time) is included in some tables as a measure of real-time performance, calculated as the average of three repeated trials. However, in certain experiments, the differences in inference times among methods are minimal, making the metric highly sensitive to device conditions and system fluctuations. Therefore, the Time is excluded from certain tables to avoid potential misinterpretation. Performance metrics include mAP_{50} and $mAP_{50:95}$. Average Precision (AP) serves as a comprehensive measure that combines precision (P) and recall (R), calculated as shown in (8),(9),(10),(11).

$$P = \frac{TP}{TP + FP} \tag{8}$$

Table 6 Details of training parameters

Parameter	Value
Image size	640×640
Batch size	8
Optimizer	AdamW
Initial learning rate	0.000238
Decay strategy	linearly
Decay rate	0.01
Weight decay	0.0005
Epoch	100
Warm-up epochs	3

Table 7 Results of the ablation experiments

Model	a	b	c	d	e	f	mAP_{50}	$mAP_{50:95}$	Parameters (M)	FLOPs (G)	Time (ms)
base							**99.0%**	**96.2%**	3.01	8.13	2.7
model 1	✓						98.5%	95.6%	1.87	7.17	2.6
model 2	✓	✓					98.7%	95.9%	1.61	4.51	2.4
model 3	✓	✓	✓				98.2%	95.3%	1.38	3.92	2.3
model 4	✓	✓	✓	✓			98.5%	95.6%	1.38	3.92	2.4
model 5	✓	✓	✓	✓	✓		97.4%	94.4%	**0.95**	**2.48**	**2.2**
CHDPL-Net	✓	✓	✓	✓	✓	✓	98.2%	95.4%	0.96	2.49	2.3

$$R = \frac{TP}{TP + FN} \tag{9}$$

where TP represents correctly detected targets, FP represents targets from other classes that are incorrectly detected as the class in question, and FN represents targets of the class in question that are either incorrectly detected as other classes or not detected at all.

$$AP = \int_0^1 P(R)dR \tag{10}$$

$$mAP = \frac{1}{m}\sum_{i=1}^{m} AP_i \tag{11}$$

where AP represents the average precision for each class, and mAP represents the mean of the average precision across all classes. The mAP_{50} calculated at IoU threshold of 0.5, reflecting the model's performance under relatively lenient matching conditions. The $mAP_{50:95}$ calculated by averaging the mAP values across multiple IoU thresholds, starting from 0.5 and increasing to 0.95 in steps of 0.05. This provides a more stringent evaluation standard.

4.3 Ablation experiments

In this study, our goal is to obtain a lightweight CHDP detection model, CHDPL-Net, by improving the original YOLOv8 model. To clarify the impact of each improvement on the network, we conducted ablation experiments using YOLOv8n as the basemodel, where a to f correspond to Sections 3.3.1 to 3.3.6, respectively.

The results are shown in Table 7. First, by adjusting the network scaling factors, we reduced the Parameters and FLOPs by 37.9% and 11.8%, respectively, while mAP_{50} and $mAP_{50:95}$ only decreased by 0.5% and 0.6%. Subsequently, we optimized the Neck and Head structures, reducing the Parameters and FLOPs by 13.9% and 37.1%, respectively, while mAP_{50} and $mAP_{50:95}$ actually increased by 0.2% and 0.3%, demonstrates that our new structure achieves lightweight design while avoid being affected by the small target detection head. To replace the costly CBS downsampling, the proposed RDown further reduced the Parameters and FLOPs by 14.3% and 13.1%, with mAP_{50} and $mAP_{50:95}$ decreasing by only 0.5% and 0.6%. The introduction of DySample slightly increased the Parameters and FLOPs but improved both mAP_{50} and $mAP_{50:95}$ by 0.3%. For further lightweighting, improvements to SPPF, C2F, and Bottleneck reduced the model's Parameters and FLOPs by 31.2% and 36.7%, respectively, but led to a more significant decrease in mAP_{50} and $mAP_{50:95}$, by 1.1% and 1.2%. To mitigate the impact of these changes on the model, we proposed and integrated the EHA lightweight attention mechanism into the tail of the C2FGA, resulting in increases of 0.8% and 1.0% in mAP_{50} and $mAP_{50:95}$, respectively. Ultimately, compared to YOLOv8n, the proposed CHDPL-Net achieved a 68.1%

Table 8 Results of comparison experiments with different scaling factors

Model	w	d	r	mAP_{50}	mAP_{50}	Parameters (M)	FLOPs (G)	Time (ms)
YOLOv8n	0.25	0.33	2.0	99.0%	96.2%	3.01	8.13	2.7
YOLOv8s	0.50	0.33	2.0	99.4%	**96.8%**	11.14	28.52	5.2
YOLOv8m	0.75	0.67	1.5	99.2%	96.3%	25.86	78.81	11.3
YOLOv8l	1.00	1.00	1.0	**99.5%**	96.6%	43.64	164.98	18.7
YOLOv8x	1.25	1.00	1.0	99.0%	96.2%	68.16	257.59	29.4
YOLOv8p	0.125	0.33	2.0	95.2%	92.0%	**0.94**	**2.89**	**2.0**
YOLOv8t (Ours)	0.25	0.33	1.0	98.5%	95.6%	1.87	7.17	2.6

reduction in parameters and a 69.4% reduction in FLOPs, with only a 0.8% decrease in both mAP_{50} and $mAP_{50:95}$. These results prove the effectiveness of the modifications we have implemented. Moreover, the overall decreasing trend in Time also indicates that our method can improve real-time performance to some extent.

4.4 Comparative experiments

To further demonstrate the effectiveness of the proposed improvements, we conducted comparative experiments to evaluate the impact of different network scaling factors, various downsampling layers, and different attention mechanisms. Additionally, we compared our model with other mainstream object detection models.

4.4.1 Different scaling factors

We conducted a comparative experiment among the scaling factors inherent in YOLOv8, the YOLOv8p proposed by Liu et al. (2024), and YOLOv8t.

As shown in Table 8, when moving from YOLOv8n to YOLOv8s, although mAP_{50} and $mAP_{50:95}$ increased by 0.4% and 0.6%, the Parameters, FLOPs, and Time increased by 270.1%, 250.8%, and 92.6%. The model parameters have already overflowed, and larger model sizes may even lead to overfitting, causing a decrease in mAP_{50} and $mAP_{50:95}$. Although YOLOv8p achieves the best performance in terms of Parameters, FLOPs, and Time, the decrease in mAP_{50} and $mAP_{50:95}$ is also significant. This is because YOLOv8p halves most of the channel numbers based on YOLOv8n, which has a considerable impact on shallow features and lower efficiency in pruning these layers. In contrast, the proposed YOLOv8t halves the channel numbers in deeper features, as deeper features contain more redundant channels and are more efficiently pruned. As a result, YOLOv8t manages to reduce Parameters and FLOPs by 37.9% and 11.8%, respectively, while only decreasing mAP_{50} and $mAP_{50:95}$ by 0.5% and 0.6%.

4.4.2 Different downsampling

We compared CBS, Adown (Wang et al. 2024), LDown (Dang et al. 2025), and RDown in CHDPL-Net.

As shown in Table 9, CBS exhibits the best detection performance, but it also has the highest Parameters and FLOPs. Adown from YOLOv9 and LDown can reduce Parameters and FLOPs by approximately 28.3% and 27.6%, respectively, but they result in relatively larger drops in mAP_{50} and $mAP_{50:95}$. In contrast, our RDown achieves reductions of 20.0% and 19.2% in Parameters and FLOPs, respectively, with minimal impact on mAP_{50} and $mAP_{50:95}$, indicating that RDown has a higher parameter efficiency.

Table 9 Results of comparison experiments with different downsampling

Downsampling	mAP_{50}	$mAP_{50:95}$	Parameters (M)	FLOPs (G)
CBS	**98.4%**	**95.5%**	1.20	3.08
ADown	98.0%	95.0%	**0.86**	2.24
LDown	97.9%	95.0%	**0.86**	**2.23**
RDown (Ours)	98.2%	95.4%	0.96	2.49

4.4.3 Different attention mechanisms

The proposed attention mechanism EHA plays a crucial role in the identification of CHDP, which is sensitive to color and texture. To validate the efficiency of EHA in our model, we compared it with several classic lightweight attention mechanisms, including SE (Hu et al. 2018), SA (Vaswani et al. 2017), CBAM (Woo et al. 2018), EMA (Wang et al. 2020), CA (Hou et al. 2021), and SimAM (Yang et al. 2021). The results are shown in Table 10.

Among various attention mechanisms, SE outperforms SA by 0.4% and 0.2% in mAP_{50} and $mAP_{50:95}$, respectively, but it also significantly increases Parameters and FLOPs. ECA, as a lightweight improvement of SE, performs slightly worse than SE in terms of mAP_{50} and $mAP_{50:95}$,but it is better than SA. However, the Parameters and FLOPs are even lower than those of SA. Although CA exhibits better performance, it has the highest Parameters and FLOPs, which is not conducive to lightweight deployment. SimAM, although lightweight, only slightly outperforms SA in performance. The EHA, based on ECA and SA, surpasses CBAM (which

Table 10 Results of comparison experiments with different attention mechanisms

Attention	mAP_{50}	$mAP_{50:95}$	Parameters (M)	FLOPs (G)
SE	98.0%	95.1%	0.966	2.498
SA	97.6%	94.9%	0.958	2.487
CBAM	98.1%	95.2%	0.966	2.505
ECA	97.9%	95.0%	**0.957**	2.486
CA	**98.3%**	95.3%	0.973	2.522
SimAM	97.8%	95.0%	0.958	**2.479**
EHA (Ours)	98.2%	**95.4%**	0.958	2.491

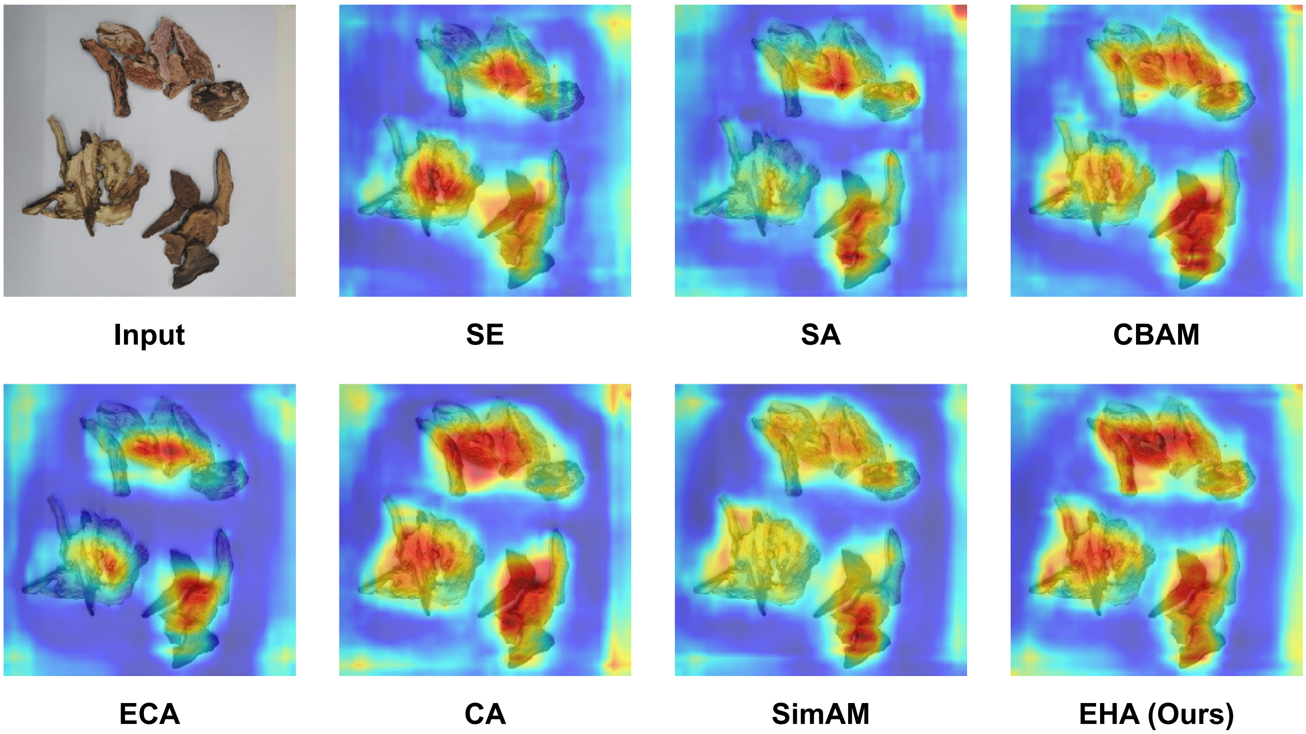


Fig. 16 Heatmaps of different attention mechanisms

concatenates SE and SA) in performance while having fewer Parameters and FLOPs, demonstrating the efficiency of EHA in our model.

The superior performance of EHA can be attributed to its parallel dual-branch structure, which achieves complementary enhancement of channel and spatial information. This improves the model's perception of color and texture features while maintaining low computational complexity. To more intuitively illustrate the effects of different attention mechanisms, we used activation heatmaps for comparison, as shown in Fig. 16. Overall, SA and SimAM performed the worst, showing insufficient sensitivity to the left-side targets. CA and EHA performed the best, exhibiting high sensitivity to each target. ECA performed slightly worse than SE, with reduced parameters leading to a smaller perception range. CBAM, which combines SE and SA, was barely able to detect each target. In summary, the proposed EHA demonstrates the best parameter utilization in the model.

Table 11 Results of comparison experiments with other models

Model	mAP_{50}	$mAP_{50:95}$	Parameters (M)	FLOPs (G)	Time (ms)
YOLOv5n	97.2%	94.5%	2.51	7.11	2.4
YOLOv8n	**99.0%**	**96.2%**	3.01	8.13	2.7
YOLOv9t	98.5%	95.6%	1.98	7.64	3.8
YOLOv10n	98.2%	95.3%	2.71	8.31	2.7
YOLOv11n	98.5%	95.7%	2.59	6.35	3.4
YOLOv12n	98.6%	95.7%	2.56	6.36	4.0
Faster-YOLO-AP	97.0%	94.4%	**0.66**	**2.29**	**2.0**
CHDPL-Net	98.2%	95.4%	0.96	2.49	2.2

4.4.4 Comparison with other algorithms

To further demonstrate the superiority of our model, we compared it with the versions of YOLO v5, v8, v9, v10 (Wang et al. 2024), v11 (Khanam and Hussain 2024), and v12 (Tian et al. 2025) that have the smallest network scale factors (i.e., the smallest model configurations) among these models. Additionally, we compared our model with Faster-YOLO-AP, an improvement on YOLOv8 by Liu et al. (2024). Since the performance differences among various models in terms of mAP_{50} and $mAP_{50:95}$ are not significant, visualizing the detection results would not reveal any noticeable differences to the naked eye. Therefore, we only present a comparison of these models' results in tabular form.

As shown in Table 11. It can be seen that YOLOv5n is somewhat outdated, performing poorly in terms of mAP_{50} and $mAP_{50:95}$. In contrast, YOLOv8n has the largest Parameters and FLOPs, with unsurprisingly the highest mAP_{50} and $mAP_{50:95}$. Notably, compared to YOLOv8n, v9t obtains competitive results by using only 65.8% of the Parameters and 94.0% of the FLOPs while experiencing only 0.5% and 0.6% drops in mAP_{50} and $mAP_{50:95}$, respectively. This is mainly due to the auxiliary reversible branch design, which mitigates semantic loss during traditional multi-path feature fusion. However, a significant decline in the inference speed of v9t is observed. YOLOv10n also performed poorly in our experiments: despite having higher Parameters and FLOPs than v9t, its mAP_{50} and $mAP_{50:95}$ were worse. YOLOv11n and v12n had mediocre performance, with v12n achieving 0.4% and 0.5% drops in mAP_{50} and $mAP_{50:95}$, using 85.0%, 78.2%, and 148.1% of the Parameters, FLOPs, and Time. Faster-YOLO-AP, with only 22.0%, 28.2%, and 74.1% of the Parameters, FLOPs, and Time, experienced a significant performance drop, with mAP_{50} and $mAP_{50:95}$ decreasing by 2.0% and 1.8%, respectively, due to an extreme focus on lightweighting. Our CHDPL-Net achieved excellent results with only 31.9%, 30.6%, and 81.5% of the Parameters, FLOPs, and Time, resulting in just a 0.8% drop in mAP_{50} and $mAP_{50:95}$, striking a better balance between accuracy and lightweight efficiency.

4.5 Robustness experiments

4.5.1 Experimental setting

We constructed a more challenging dataset named CHDP5H, which comprises 12 CHDP categories with a total of 528 images. This dataset incorporates 6 novel CHDP categories, increased numbers of similar categories, and more complex scenarios to further validate the robustness of CHDPL-Net. The scenario examples and specific categories are illustrated in Figs. 17 and 18 respectively. The dataset was partitioned

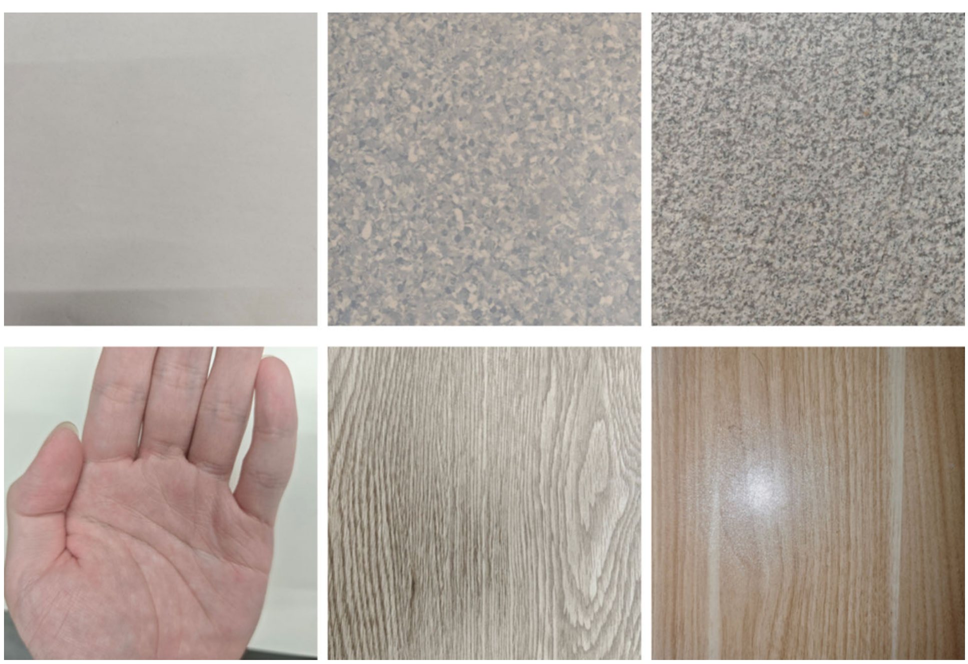


Fig. 17 Scenario examples

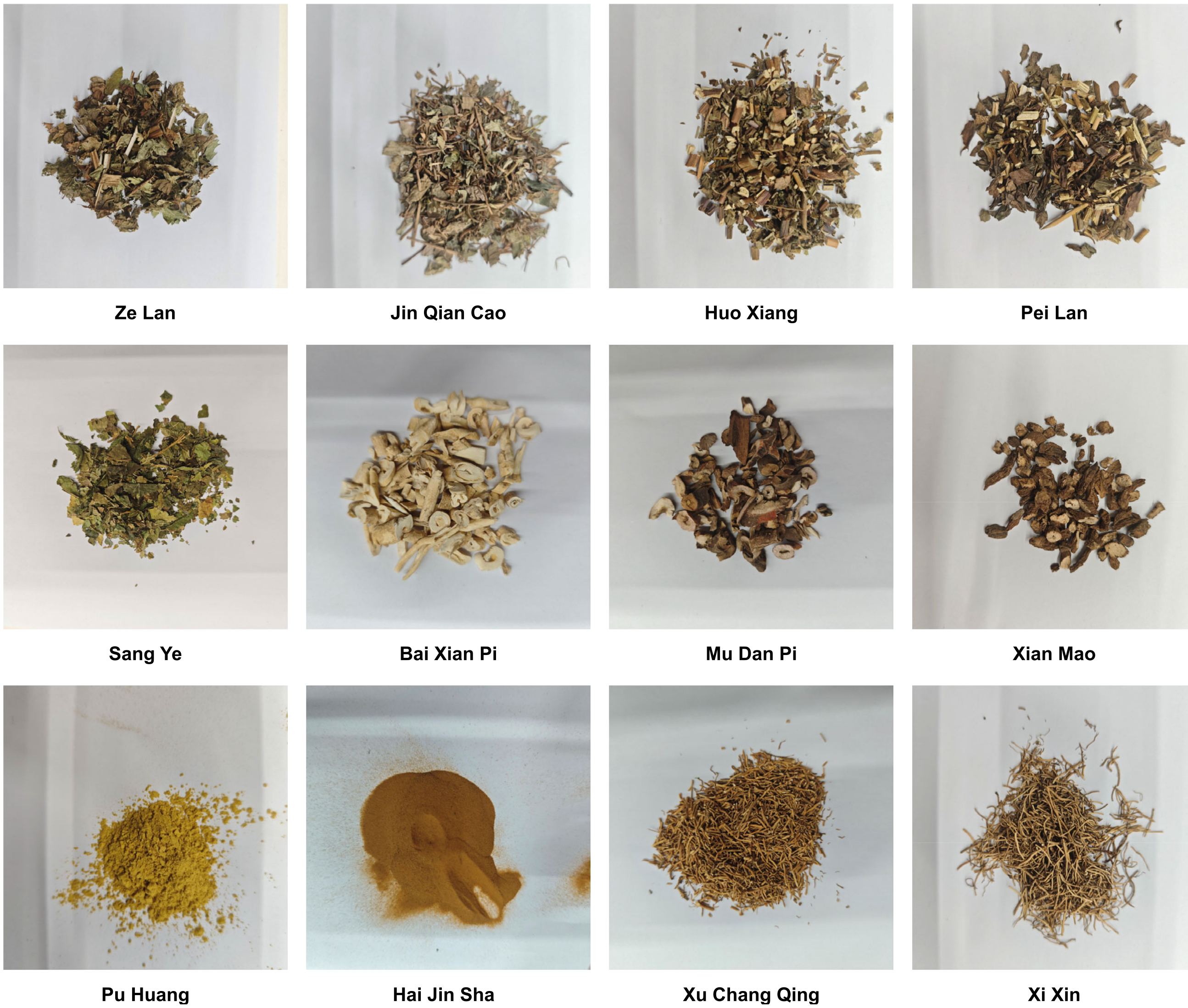


Fig. 18 Categories of CHDP

into training and testing sets at a 3:1 ratio, with the initial learning rate adjusted to 0.000625 based on the quantity of category.

4.5.2 Results analysis

The results are shown in Table 12. The mAP_{50} reached 97.2%, with a decrease of only 1.0%, indicating that the proposed model has high robustness and can maintain high accuracy even when facing more similar CHDPs and complex scenarios. There was a more noticeable decline in $mAP_{50:95}$, at 5.7%, primarily due to the influence of Huo Xiang, Xu Chang Qing, Xi Xin, Pu Huang, and Hai Jin Sha. For these CHDPs, some images encountered difficulties in selecting appropriate boundaries during data collection, making it challenging for the model to accurately determine the specific boundaries and achieve high-precision localization, as shown in Fig. 19.

The confusion matrix is shown in Fig. 20. Combined with the results in Table 12, it can be observed that prediction errors mainly occur among four CHDPs: Ze Lan, Huo Xiang, Jin Qian Cao, and Pei Lan. This is primarily due to their similar morphologies—each consists of stems with leaves—and their comparable color characteristics. Among these, Huo Xiang and Pei Lan are the most visually similar. However, Huo Xiang typically contains many small, fragmented leaves,

Table 12 Results of robustness experimental

CHDP	Precision	Recall	AP_{50}	$AP_{50:95}$
Sang Ye	93.8%	93.8%	98.8%	91.7%
Ze Lan	72.4%	81.2%	93.9%	91.2%
Huo Xiang	99.5%	78.3%	96.9%	87.7%
Jin Qian Cao	85.7%	93.8%	95.7%	89.9%
Pei Lan	92.7%	79.9%	92.9%	84.9%
Xu Chang Qing	94.0%	99.5%	96.6%	84.1%
Xi Xin	90.0%	93.8%	97.8%	89.7%
Xian Mao	99.5%	74.4%	97.8%	93.6%
Mu Dan Pi	84.3%	93.8%	97.5%	92.5%
Bai Xian Pi	97.3%	99.5%	99.5%	97.7%
Pu Huang	97.3%	99.5%	99.5%	80.5%
Hai Jin Sha	97.2%	99.5%	99.5%	92.7%
All	92.1%	90.7%	97.2%	89.7%

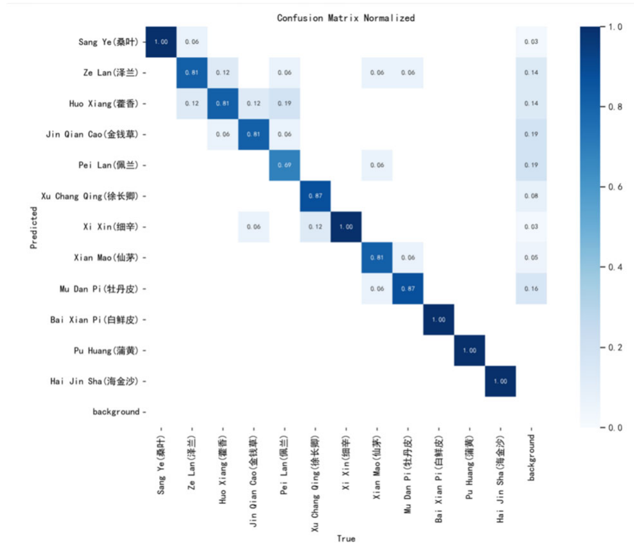


Fig. 20 Confusion matrix

while Pei Lan is characterized by its darker leaf coloration. Thanks to the EHA attention mechanism's ability to effectively capture and integrate both texture and color features, Huo Xiang and Pei Lan still achieve mAP_{50} scores of 96.9% and 92.9%, respectively, demonstrating strong recognition performance, as illustrated in Fig. 21. Ze Lan, due to its relatively abundant foliage, has a low probability of being incorrectly identified as Sang Ye.

Xu Chang Qing and Xi Xin also form a pair of highly similar CHDPs. Among them, Xu Chang Qing is characterized by its more fragmented appearance and deeper coloration. Our model demonstrates strong discrimination capability between the two, achieving mAP_{50} scores of 96.6% and 97.8%, respectively. Fig. 22 shows an example of incorrect prediction, primarily due to the Xi Xin samples in the upper-

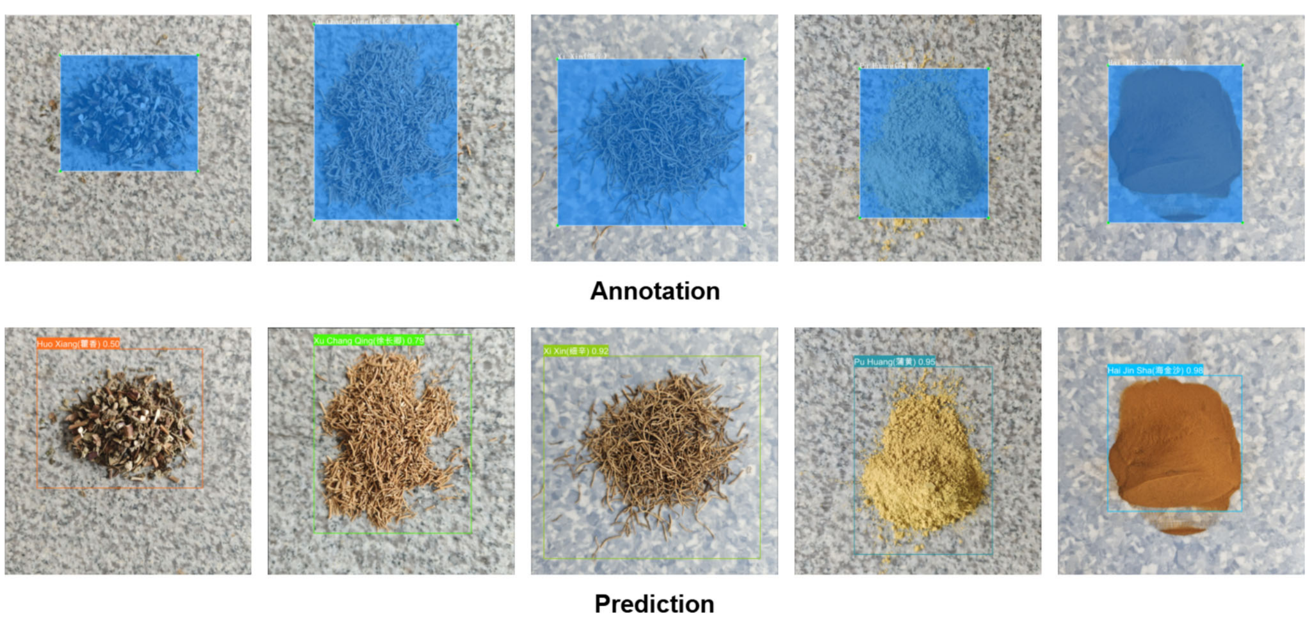


Fig. 19 Annotation-Prediction divergence

Fig. 21 Predictions of Hou Xiang and Pei Lan

right corner were captured with low clarity, which reduces the distinctiveness of their features. This issue is further aggravated when the images are resized to 640×640 pixels for input into the model.

Despite the high morphological similarity between Mu Dan Pi and Bai Xian Pi, they are relatively easy for the model to distinguish, mainly due to their significant color differences. In contrast, Xian Mao is more likely to be confused with Mu Dan Pi, as shown in Fig. 23. This is because Xian Mao shares a similar color with Mu Dan Pi, and its morphological differences are small—especially when the samples are stacked, further reducing visual distinction. Nevertheless, the mAP_{50} scores for Mu Dan Pi and Xian Mao remain high at 97.5% and 97.8%, respectively. In addition, although both Pu Huang and Hai Jin Sha appear in powdered form, they can still be effectively distinguished due to their clearly different colors, as shown in Fig. 24.

Overall, the CHDPL-Net demonstrates excellent detection performance. Despite the relatively small dataset and the presence of more similar CHDP categories along with complex real-world scenarios, the model maintains a high level of recognition accuracy. It exhibits strong robustness

Fig. 22 Predictions of Xu Chang Qing and Xi Xin

Fig. 23 Predictions of Mu Dan Pi and Xian Mao

Fig. 24 Predictions of Pu Huang and Hai Jin Sha

and is capable of meeting practical detection requirements to a considerable extent.

5 Conclusion

In response to the demands of digital transformation in TCM, this study proposes a lightweight network named CHDPL-Net for CHDP detection. This method proposes new network scale factors based on YOLOv8 to reduce channel redundancy in deep features and optimizes the Neck and Head structures to accommodate the characteristics of CHDP, which primarily involves medium to large targets. Additionally, the proposed RDown module minimizes parameters and computational costs, and DySample is adopted to enhance fine-grained feature reconstruction. Finally, GhostConv is introduced to lightweight the C2F and SPPF modules, with the EHA integrated at the C2F tail to strengthen color-texture feature perception. Experimental results on the self-constructed large-scale dataset CHDP4K demonstrate that the improved model achieves mAP_{50} and $mAP_{50:95}$ scores of 98.2% and 95.4%, respectively, utilizing only 31.9% of Parameters and 30.6% FLOPs compared to YOLOv8. Furthermore, its robustness is validated on the more complex self-built dataset CHDP5H. These results demonstrate that the proposed solution significantly reduces computational overhead while maintaining high detection accuracy, highlighting its practical applicability in real-world scenarios.

This research also has certain limitations. Although robustness experiments were conducted on a dataset with more complex scenes and multiple similar categories, more complex scenarios and additional similar classes may appear in practical applications. Therefore, future work will explore a dynamic incremental update framework based on continual learning to address these challenges. Additionally, this study is limited to the identification of CHDP categories, while the medicinal properties and quality of CHDP are also key factors that need attention throughout the entire process. In TCM practice, the color and texture of CHDP are important indicators of its medicinal properties and processing quality. For example, variations in color often reflect differences in efficacy, while subtle details in texture are closely related to the origin or preparation method. Thus, how to assess the medicinal properties and quality of CHDP will also be a focus of our future research.

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Data Availability Data will be made available on reasonable request.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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